

Lab 10 Solutions(Grp 9-16)

Surge prices :

```
class Metro:
    def __init__(self, line, fare):
        self.line = line
        self.fare = fare

    def surgePrice(self):
        sp = 0
        if self.line == "Red":
            sp = 1.5 * self.fare
        if self.line == "Blue":
            sp = 2 * self.fare

        return sp

l = input()
f = int(input())

mdetail = Metro(l, f)

print(f'{mdetail.surgePrice():.1f}')
```

Cuboids :

```
class Cuboid:
    def __init__(self,x,y,z):
        self.l,self.b,self.h = x,y,z
    def getVolume(self):
        return self.l *self.b*self.h
    def getTotalSurfaceArea(self):
        return 2*(self.l*self.b + self.l * self.h + self.b * self.h)
    def hasSameVolume(self, obj):
        return "YES" if self.getVolume() == obj.getVolume() else "NO";

if __name__ == "__main__":
```

```

n = int(input())
cuboid_obj = []
ans = ""
for i in range(n):
    inp = input().split(" ")
    obj = Cuboid(int(inp[0]),int(inp[1]),int(inp[2]))
    cuboid_obj.append(obj)
q = int(input())
for i in range(q):
    inp = input().split(" ")
    t = int(inp[0])
    if t==1:
        index = int(inp[1])
        ans += str(cuboid_obj[index].getVolume())
    elif t==2:
        index = int(inp[1])
        ans += str(cuboid_obj[index].getTotalSurfaceArea())
    else :
        x = cuboid_obj[int(inp[1])]
        y = cuboid_obj[int(inp[2])]
        ans += x.hasSameVolume(y)
    ans += "\n"
print(ans)

```

Report Card :

```

class reportcard:
    def __init__(self,studentname,studentid,marks):
        self.studentname=studentname
        self.studentid=studentid
        self.marks=int(sum(marks)/len(marks))
    def getfmarks(self):
        print(self.studentname,self.studentid,self.marks,end=' ')
    def getgrade(self):
        if self.marks>90:
            print('A')
        elif self.marks>80 and self.marks<=90:
            print('B')
        elif self.marks>70 and self.marks<=80:
            print('C')
        else:
            print('D')

```

```

Studentname=input()
studentid=int(input())
marks=[int(i) for i in input().split()]
a=reportcard(Studentname,studentid,marks)
a.getfmarks()
a.getgrade()

```

Number game 28:

```

temp = input().split()
a = int(temp[0])
b = int(temp[1])
class Number:
    def __init__(self, a, b):
        self.a = a
        self.b = b
    def AM():
        return((a+b)/2)
    def GM():
        return((a*b)**(1/2))
    def HM():
        return(2/((1/a)+(1/b)))
print(*[Number.AM(), Number.GM(), Number.HM()])

```

Dictionary Search 1 :

```

import math
class Point:
    def __init__(self, x,y):
        self.x = x
        self.y = y
    def x_co(self):
        for i in self.x:
            print(i)
    def distance_(self):
        return(math.dist(self.y, self.x))

```

```

x = [int(i) for i in input().split()]
y = [int(j) for j in input().split()]

```

```

coords = Point(x,y)

```

```
coords.x_co()  
print(coords.distance_())
```